

Shivani R. Shukla, PhD

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CAREER OBJECTIVE

I build in vitro platforms that read out glial and neuroinflammatory state electrically, at single-cell resolution, to study the mechanisms that drive electrode-tissue interface degradation.

EDUCATION

University of California - San Diego, PhD in Bioengineering	September 2020 - March 2025
Materials Research Science and Engineering Center (MRSEC) Graduate Research Fellowship	2024
Merkin Graduate Fellowship at UC San Diego	2023
Interfaces Graduate Training Program at UC San Diego	2022
National Science Foundation Graduate Research Fellowship	2020
Carnegie Mellon University, Bachelor of Science, Materials Science & Engineering	September 2016 - May 2020
Additional Major in Biomedical Engineering	
University and College Honors	
BME Undergraduate Research Award	2020
Buildium Women in Technology Scholarship	2018

WORK EXPERIENCE

Postdoctoral Fellow | Biomimetrics

Chan Zuckerberg Biohub, Chicago

September 2025 - Present

- Novel fabrication methods for scalable nanoelectrode arrays using machine learning-guided two-photon polymerization, reactive ion etching, electron beam lithography, and interference lithography
- Topological routing for scalable, cross-platform bioelectronic design
- Flood filling networks for mixed hiPSC-derived brain cell cultures to assess neuroinflammation states on both immunofluorescent and electron microscopy datasets
- Develop a generalized Hodgkin Huxley model combined with pharmacological models for glial cell perturbations and functional assay development
- Designed and executed cell-based assays (MSD ELISA, high content imaging) to investigate neuroinflammatory mechanisms for understanding of disease pathways
- Creating multiphysics simulations for optimizing cell-electrode coupling in bioelectronic interfaces
- Design substrate-bound proximity assay using click chemistry for ion channel and GPCR sampling

RESEARCH EXPERIENCE

Graduate Researcher | Thesis Advisors: Drs. Zeinab Jahed and Gert Cauwenberghs | Dept of Bioengineering

University of California, San Diego

August 2021 -

March 2025

- *Thesis: Multiscale Intracellular-like Recording Using Nanopillar Electrode Arrays*
- Sub- and supra-threshold intracellular electrophysiology and signal analysis of primary and induced pluripotent stem cell derived (iPSC) 2D and 3D neuronal networks and cardiomyocyte monolayers using nanopillar electrode arrays
- Developed tunable and scalable nanofabrication pipeline for semi-transparent, bidirectional, nanostructured electrode arrays using multi-layer photolithography and two-step etching
- Simultaneous intracellular electrophysiology/electrochemical sensing of neuronal networks using nano- and microelectrode arrays
- Differentiation of iPSC-derived neurons including healthy, diseased, and aged models
- Cell culture optimization and interfacing of iPSC-derived cardiomyocyte monolayers and 3D-printed tissues
- Performed conventional spike sorting and multimodal, multiscale analysis on intracellular recordings in Python
- Mentored several graduate and undergraduate students on cell culture, electrophysiology, and nanofabrication

- Developed extensive grant, fellowship, and peer review writing and communication skills across several fields

Merkin Graduate Fellowship | PI: Francesca Santoro | Institute of Bioelectronics

Forschungszentrum Juelich, Germany

January - August 2024

- Simultaneous patch clamp and nano/microelectrode array recording of neuronal networks
- Neuronal guidance and synaptic patterning using immunofluorescence and subcellular calcium imaging

Research Rotation | PI: Dr. Ester Kwon | Department of Bioengineering

University of California, San Diego,

September 2020 - August 2021

- Developed multiplexed lateral flow assays to detect protease activity after elevation post-TBI in mice
- Developed behavioral testing protocol in mice using the Rotarod and pre/post-operative care to stratify TBI severity using Controlled Cortical Impact (CCI) model
- Performed CCI surgeries on mice as well as tail vein, intraperitoneal, and intranasal injections for various treatments
- Mentored 5 high school and undergraduate students on LFA construction and design

SURF Fellow | PI: Dr. Tzahi Cohen-Karni | Department of Biomedical Engineering

Carnegie Mellon University, Pittsburgh, Pennsylvania

September 2017 - May 2020

- Synthesized 3 different nano- and micro-topographic graphene-based surfaces (GBS) using Plasma-Enhanced CVD
- Quantified focal adhesions & differentiation of neurons and C2C12 muscle cells on GBS, gold, and glass
- Designed microelectrode arrays (MEAs) in AutoCad for electrophysiology of 3D organoid cardiac and neural tissues
- Performed confocal z-stack imaging of organoids encapsulated inside of self-rolling MEAs

ISURF Fellow | PI: Dr. Avi Schroeder | Department of Chemical Engineering

Technion – Israel Institute of Technology, Haifa, Israel

May - August 2018

- Developed chemical ATP synthesis method for artificial mitochondria in tumor-bound anti-cancer synthetic cells

Nakatani-RIES Fellow | PI: Dr. Hisanori Shinohara | Department of Chemistry

Nagoya University, Nagoya, Japan

June - August 2017

- Isolated high-yield, 1D transition metal chalcogenide nanowires by encapsulation inside carbon nanotubes
- Wrote English TEM Operation Manual for international visitors to the lab

LEADERSHIP AND TEAMWORK

21st Annual International Summer Leadership Academy on Bio-X: AI in Healthcare, Medicine and Biology 6/2024

- Attended lectures and discussions about grand challenges at the intersection of AI and bioengineering
- IEEE Engineering in Medicine and Biology Society (EMBS) Student Chapter @ UC San Diego, President 8/2023-present

- Professional development and mentorship from EMBS industry guests in San Diego

Neuroengineering Journal Club @ UC San Diego, Founder

August 2022 - present

- Monthly gathering of graduate students to discuss neural engineering literature, theory, and recent advancements

Industry Committee, Bioengineering Graduate Society, Co-chair

April 2022 - present

- Hosted Illumina Fireside Chat representing 5 different divisions of Illumina with 70+ attendees
- Co-chaired the 27th annual Breakfast With Industry event at UCSD with 100+ attendees

Manuscriptor, LLC, CEO

May 2020 - present

- Assist in writing English manuscripts for Japanese nanochemistry researchers
 - Toolkit development for bioelectronics researchers
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BIBLIOGRAPHY

- Shukla, S, Chang, A, Tahir, A, Khan, I, Pham A, Wang, J, & Jahed, Z. Chemically Selective Nanoelectrode Arrays for Real-Time, Parallel Neurotransmitter and Electrical Recording. *Small Science* (2026).

- Shukla, S, Walsh, C., Bansal, S., Leugering, J., Uppal, A., Jaltare, V., Cauwenberghs, C., Chalasani, S., & Jahed, Z. A Dynamic Spike Sorter for Intracellular-Like Recordings. *Advanced Materials Interfaces* (2025).
- Shukla, S, Schwartz, J.L., Walsh, C., Wong, W., Patel, V., Hsieh, Y., Onwuasoanya, C., Chen, S, Offenhäusser, A., Cauwenberghs, G., Santoro, F., Muotri, A.R., Yeo, G.W., Chalasani, S., Jahed, Z. Supra- and sub-threshold intracellular-like recording of 2D and 3D neuronal networks using nanopillar electrode arrays. *Microsystems & Nanoengineering* (2024). DOI: 10.1038/s41378-024-00817-y
- Shukla, S., Comerci, C., Suel, G., & Jahed, Z. Electrical Signaling: A universal language for inter-species and inter-kingdom communication. *Biosensors & Bioelectronics* (2024). DOI: 10.1016/j.bios.2024.116843
- Sarikhani, E, Patel, V, Li, Z, Meganathan, DP, Rahmani, K, Sadr, L, Hosseini, R, Visda, D, Shukla, S, Naghsh-Nilchi, H, Balaji, A, McMahon, G, Chen, S, Schöneberg, J, McHugh, CA, Shi, L, Jahed, Z. Engineered Nanotopographies Induce Transient Openings in the Nuclear Membrane. *Advanced Functional Materials* (2024).
- Kalmykov, A., Huang, C., Bliley, J., Shiowski, D., Tashman, J., Abdullah, A., Rastogi, S.K., Shukla, S., Mataev, E., Feinberg, A. W., Hsia, K. J., & Cohen-Karni, T. Organ-on-a-chip: Three-dimensional self-rolled biosensor array for electrical interrogations of human electrogenic spheroids. *Science Advances*, 5(8) 2019. DOI: 10.1126/sciadv.aax0729
- Nagata, M. & Shukla, S., Nakanishi, Y., Liu, Z., Lin, Y., Shiga, T., Nakamura, Y., Koyama, T., Kishida, H., Inoue, T., Kanda, N., Ohno, S., Sakagawa, Y., Suenaga, K., & Shinohara, H. Isolation of Single-Wired Transition-Metal Monochalcogenides by Carbon Nanotubes. *ACS Nano Letters*, 19(8) 2019. DOI: 10.1021/acs.nanolett.8b05074

PRESENTATIONS

Poster Chicago BioEngineering Conference, 2026, Chicago, Illinois	September 2026
<i>Multiscale, Intracellular Recording from Neuronal Networks Using Nanopillar Electrode Arrays</i>	
Talk Materials Research Society, 2024 Annual Meeting, Boston, Massachusetts	December 2024
<i>Supra- and sub-threshold intracellular electrophysiology of neuronal networks using nanoelectrode arrays</i>	
Poster Biophysical Society, 2023 Annual Meeting, San Diego, California	February 2023
<i>A versatile nano-scale platform for intracellular electrophysiology of electrogenic cells</i>	
Talk Society for Neuroscience, 2022 Annual Meeting, San Diego, California	November 2022
<i>Morphology and Electrophysiology of Neuron-Like PC12 Adherent Cells on Nanopillar Electrode Arrays</i>	
Talk Biomedical Engineering Society 2022 Annual Meeting, San Antonio, Texas	October 2022
<i>A versatile nanoelectrode platform for electrical recording of diverse cell types</i>	
Talk Biomedical Engineering Society 2019 Annual Meeting, Philadelphia, Pennsylvania	October 2019
<i>Myogenic Differentiation on Conductive, Nanocarbon Substrates</i>	
Talk Nakatani-RIES Research Update at Tokyo University, Tokyo, Japan	August 2017
<i>Synthesizing TMD Nanowires inside Carbon Nanotubes "Nano-test-tubes"</i>	

TECHNICAL SKILLS

In Vivo Techniques: Mouse surgeries and behavioral testing (Rotarod, CCI model), Rodent brain tissue preparation and analysis, TBI model implementation and assessment

In Vitro Techniques: Primary neuronal culture and iPSC-derived neuronal differentiation, 2D and 3D neuronal network culture systems, Cardiomyocyte monolayer and cardiac organoid culture, Patch clamp electrophysiology and calcium imaging, Microelectrode array recordings (nano/micro scale), Simultaneous electrophysiology/electrochemical sensing, Confocal Immunostaining, Plasmid construction, amplification, and transfection, ELISA, cytokine arrays

Materials Characterization & Synthesis: Nanofabrication and nanopillar electrode array fabrication, Transmission/scanning electron microscopy (TEM/SEM), Raman spectroscopy and confocal z-stack imaging, Wet transfer of graphene and graphene-based surface synthesis, Click chemistry and nanomaterials synthesis, Electrochemical impedance spectroscopy, Lateral flow assay construction and multiplexed sensing, two-photon polymerization

Signal Processing & Data Analysis: Multi-scale neural signal analysis (single-cell to network), Machine learning for spike sorting and feature extraction, Multi-modal data integration and visualization

Programming & Software: MATLAB, Python, C; Data visualization: Gephi, D3.js, ImageJ; CAD design for biomedical devices; Java, JavaScript, KiCAD, PyTorch